

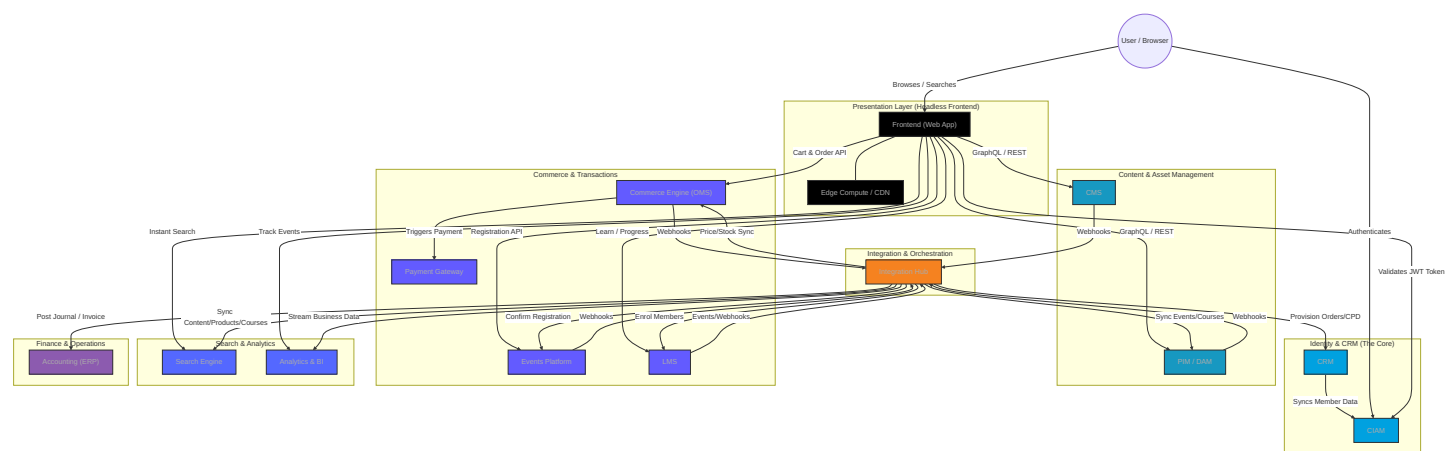
Headless Composable MACH Architecture (Abstracted Target)

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This document provides a comprehensive overview of the generic, headless, composable architecture design for the platform. It is split into two sections: the **Abstract Target Architecture** (the full target ecosystem) and the **Current Implementation Status** (what is currently integrated).

1. Abstract Target Architecture

This diagram outlines the complete ecosystem of headless services. It highlights the decoupled nature of composable services and the central role of **CIAM** and **CRM** in managing user identities and entitlements, while integrating **Accounting** and **Analytics** downstream.



Key Components:

- **Presentation:** Decoupled framework-based frontend (e.g. Next.js) hosted on a CDN / Edge Hosting platform.

- **Identity & CRM: CIAM** serves as the central identity platform, handling secure user authentication, session management, and issuing JWT tokens at the edge. **CRM** remains the core customer relationship manager and single source of truth for member records and master data, syncing status to the CIAM platform.
- **Integration: Integration Hub** (IPaaS) acts as the central integration engine, managing all backend synchronization, event indexing, and back-office automated workflows.
- **PIM & DAM: PIM / DAM** manages structured product data, including books, memberships, and digital assets.
- **CMS: CMS** manages marketing pages, news, and editorial content.
- **Commerce & Transactions: Commerce Engine** handles the shopping cart, order management (OMS), and fulfillment logic.
- **Payment Gateway: Payment Gateway** handles secure payment processing.
- **Search: Search Engine** provides high-performance, low-latency search across both content and products.
- **Learning Management System: LMS** serves as the learning management system, delivering educational content and tracking user progress.
- **Events Management: Events Platform** serves as the events management system, handling event registration and attendee management.
- **Accounting: Accounting (ERP)** handles financial ledger entries, invoicing, reconciliation, tax compliance, and overall financial reporting.
- **Reporting & Analytics: Analytics & BI** platform aggregates event streams and transaction data to perform business intelligence, user behavior analysis, and executive reporting.

Integration Reliability & Orchestration Strategy

- **Delivery Guarantees:** The Integration Hub implements at-least-once delivery for webhook events.
- **Idempotency:** All downstream sync consumers (CRM, Accounting, Search Engine) enforce idempotency checks using unique event IDs or transaction hash signatures.
- **Retry Policy:** Transient network or system errors trigger exponential backoff with jitter (max 5 retries).
- **Failure Handling:** Undeliverable messages after maximum retries are routed to a Dead-Letter Queue (DLQ) for alerting, monitoring, and manual operations intervention.

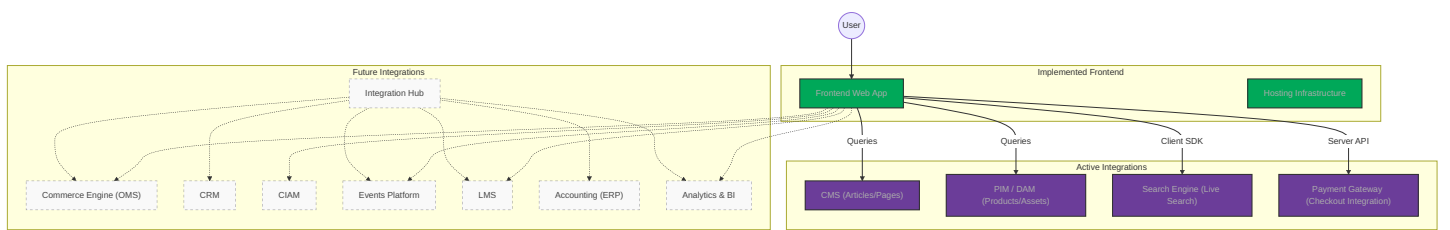
Auth & Identity Lifecycle Management

- **CRM-to-CIAM Sync:** Profile changes and membership statuses updated in the CRM are pushed to the CIAM in near real-time via webhooks (< 5s latency). A nightly batch job reconciles out-of-sync profiles.
- **Token Lifecycle:** Access tokens (JWT) carry a 15-minute lifespan. Refresh tokens are secured via `httpOnly` secure cookies with a 14-day validity.

- **Session Revocation:** A backchannel logout API allows global session invalidation directly from the CRM/CIAM interface, forcing client apps to perform re-authentication.

2. Current Implementation Status

This diagram highlights the generic roles that have been implemented and connected in the current codebase. Active integrations are highlighted in color, while future/planned integrations (including Finance, CRM, and Analytics systems) are shown in grayscale.



Component Status Matrix:

Component	Status	Notes
Frontend Web App	Implemented	Built with App Router; UI and pages styled.
Hosting Infrastructure	Implemented	Edge routing and build pipelines configured.
CMS	Implemented	Connected via <code>src/lib/cms.ts</code> fetching utility.
PIM / DAM	Implemented	Connected via GraphQL schema queries <code>src/lib/pim.ts</code> .
Search Engine	Implemented	Connected via Search SDK in frontend components.
Payment Gateway	Implemented	Backend checkout session endpoints verified.
Commerce Engine (OMS)	In Design	Shifting cart state logic from frontend cache to OMS APIs.
CRM	In Design	Designing integration schema for membership objects.
CIAM	In Design	Finalizing authentication callbacks and edge JWT verification.
Events Platform	In Design	Coordinating webhook registrations.

Component	Status	Notes
LMS	In Design	Configuring SSO launcher schema.
Integration Hub	In Design	Mapping field transformations for primary recipes.
Accounting (ERP)	Not Started	Pending definition of chart of accounts.
Analytics & BI	Not Started	Pending definition of dashboard KPIs.

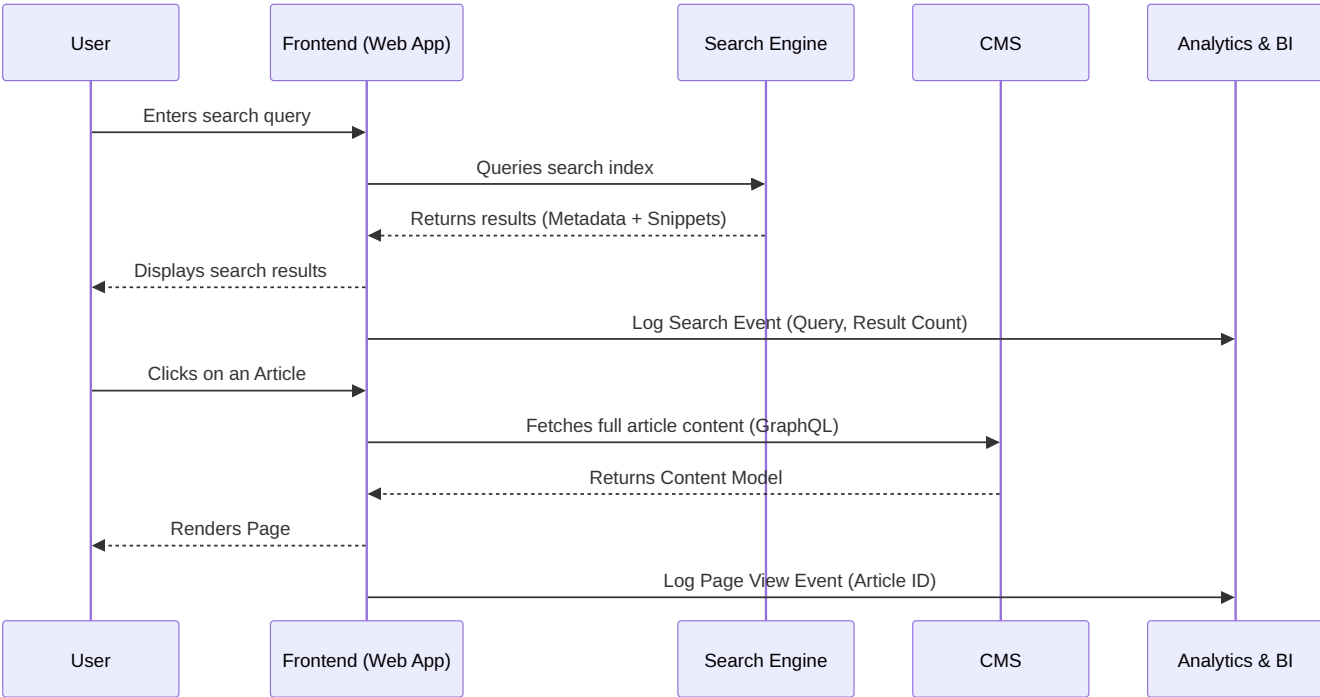
Index Sync & Degraded Mode Behavior

- **Search Index Sync:** CMS and PIM write events trigger webhook-based re-indexing. Target indexing propagation latency is < 2 minutes.
- **Degraded Mode Fallback:** If the Search Engine service becomes unavailable, the Frontend Web App gracefully falls back to displaying local search caches or query options pointing directly to the CMS/PIM database. Users are shown a temporary "search indexing lagging" notice if the index is reporting delays.

3. User Journeys & Interaction Flows

A. Visitor Searches for Information

A visitor searches the site for resources or news.

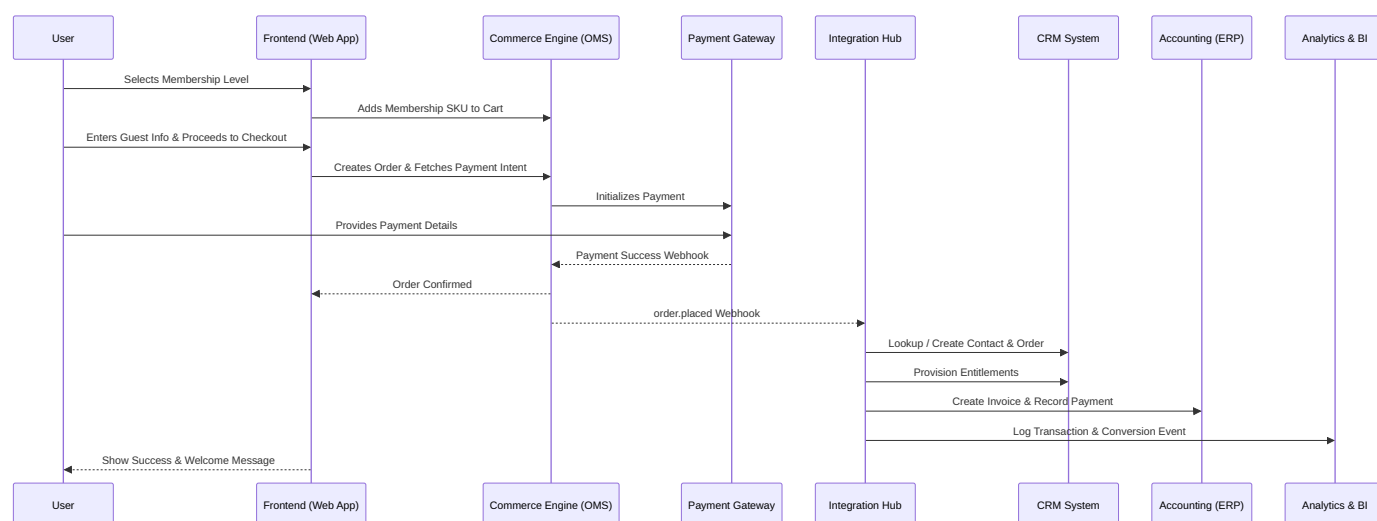


Exception Paths:

- **Zero Results:** The Frontend displays popular topics and suggests spelling corrections or alternative tags.
- **Search Timeout / Service Down:** The search bar falls back to direct, uncached queries against the CMS database, showing a degraded-search indicator.

B. Visitor Purchases Membership

A visitor signs up for a platform membership to access gated benefits.

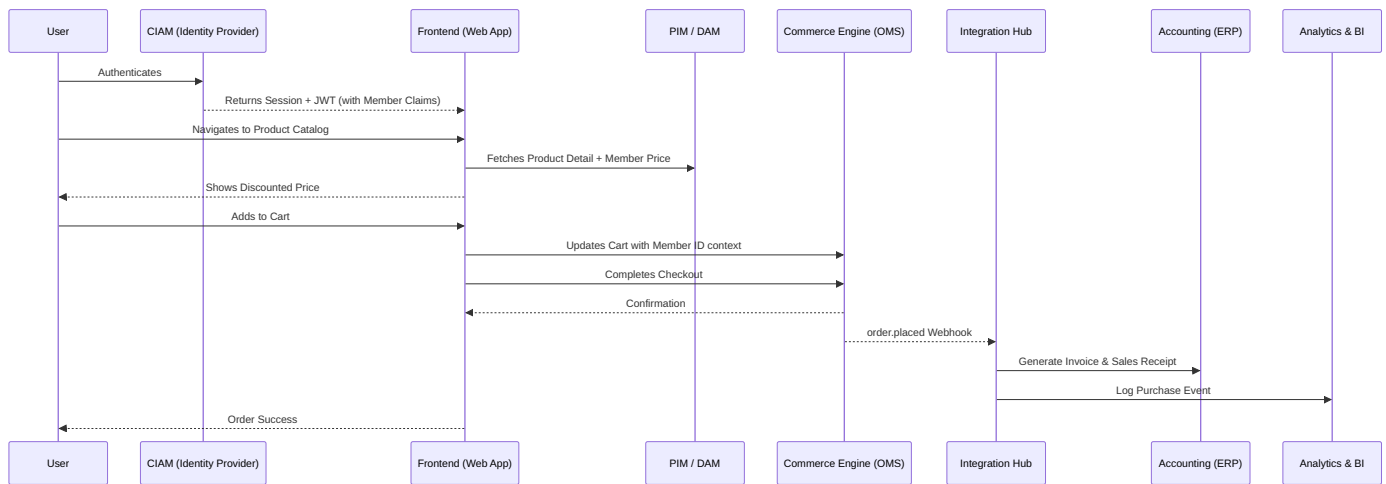


Exception Paths:

- **Payment Decline:** The Payment Gateway returns a decline status. The checkout UI preserves cart details, displays a retry option with descriptive feedback, and limits rapid checkout requests to prevent abuse.
- **Duplicate Webhook Delivery:** The Integration Hub checks the incoming payment/order ID signature. Duplicate events are silently discarded, preventing multiple accounts or invoices from being generated.
- **Identity Resolution:** Matching is performed against the primary email address. Duplicate contact handling follows the rules detailed in Section 4.

C. Member Buys a Product

An authenticated member purchases a physical publication using member pricing.

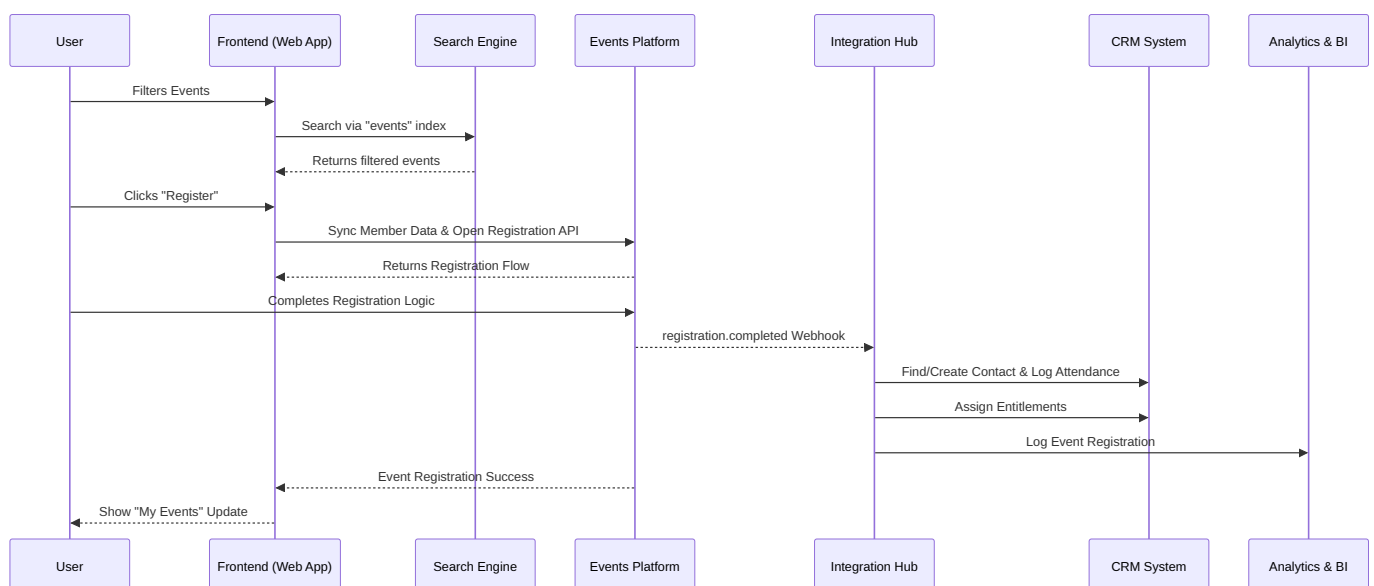


Exception Paths:

- **JWT Expiry:** If the JWT token expires during the session, the Frontend silently triggers a refresh request via the CIAM secure `httpOnly` cookie. If the user session has completely expired, the cart is cached locally and the user is redirected to re-authenticate.
- **Downstream Accounting Sync Failure:** If the Accounting ERP integration fails post-checkout, the Integration Hub stores the payload in a persistent error queue. An automated background worker attempts retries. If unsuccessful after 24 hours, it flags the transaction for manual entry. The user-facing confirmation is not impacted.

D. Member Books an Event

An existing member registers for an event or conference.

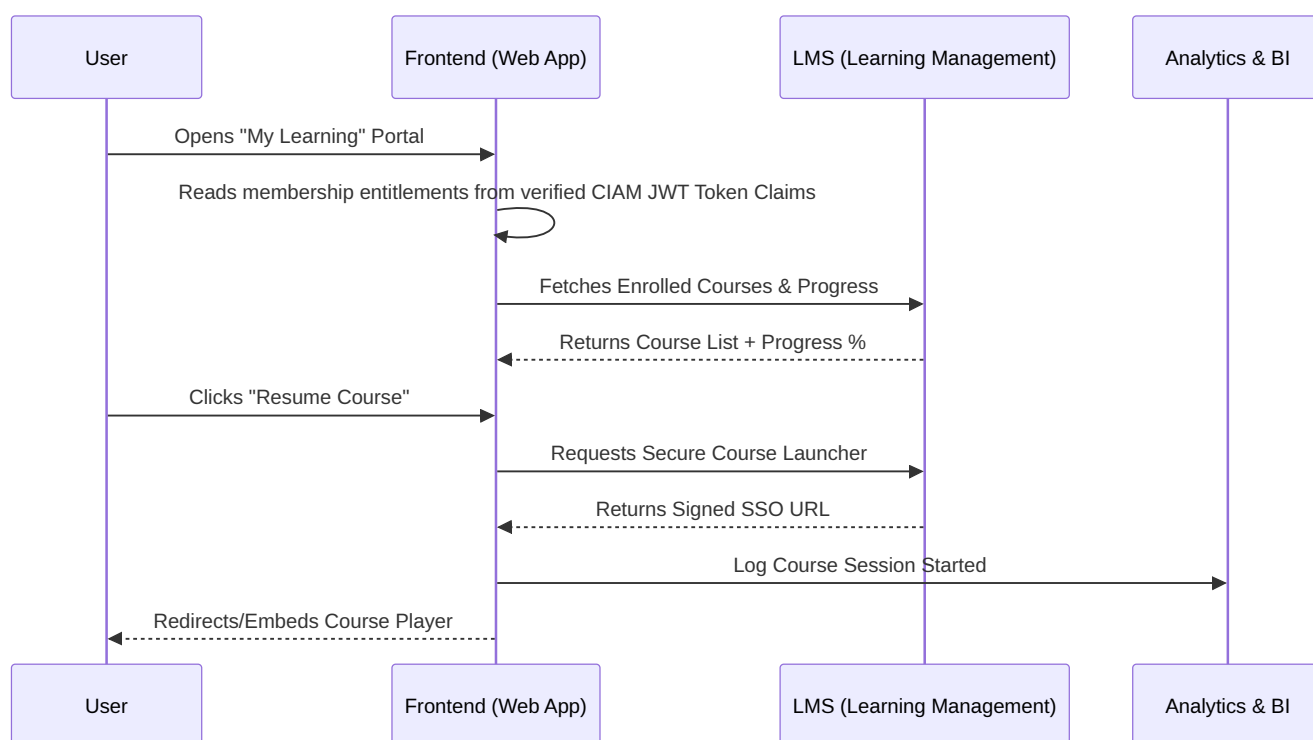


Exception Paths:

- **Sold-Out / Capacity Exceeded:** The Frontend queries event tickets in real-time. If zero tickets remain, the "Register" button is replaced by a "Join Waitlist" workflow.
- **CRM Match/Update Failure:** If the CRM cannot resolve the user during registration log syncing, the integration issues an administrative alert to CRM operations to manually link the event attendance record.

E. Member Studies a Course

A member accesses their learning dashboard to continue an online course.



Exception Paths:

- **Entitlement Sync Lag:** If a user logs in immediately after buying a membership, they might have stale JWT entitlements. The Frontend provides a "Refresh Access" button that triggers a backchannel claims refresh against the CIAM.
- **LMS SSO Gateway Downtime:** In case of LMS service failure or invalid SSO signatures, the Frontend catches the error, displays an error screen ("Course portal is currently unavailable"), and logs the event to operations.

4. Unified Identity & Data Entity Model

To resolve cross-system customer matching and avoid duplicate records, the platform relies on the following identity schema mapping:

Entity	Identifier (ID)	System of Record (SoR)	Downstream Sync Systems
Person / Contact	Contact ID	CRM	CIAM, Commerce Engine, LMS, Events, Analytics
Credentialed User	User UID	CIAM	Frontend Web App, LMS, Events
Transaction Customer	Customer ID	Commerce Engine	CRM, Payment Gateway, Accounting (ERP)
Event Registrant	Attendee ID	Events Platform	CRM (via Integration Hub)
Course Learner	Student ID	LMS	CRM (via Integration Hub)

Identity Resolution Rules:

- Primary Matching Key:** The verified email address serves as the unique identifier across all systems.
- Lookup Workflow:** When the Integration Hub receives an external webhook (e.g. checkout, event booking), it performs an email lookup in the CRM.
- Deduplication Rules:**
 - If a match is found, the transaction/activity is linked to the existing Contact ID.
 - If no match is found, the Integration Hub generates a new Contact record in the CRM.
 - If multiple duplicate email records are flagged, the transaction is associated with the most recently active account and flagged for administrative merge.

5. Non-Functional & Governance Requirements (NFRs)

Availability & Performance SLAs

- Frontend Application & CDN:** 99.9% availability target. Page load times < 1.5s for static pages.
- CIAM & Payment Gateway:** 99.99% availability targets.
- Core API Integrations (CRM, Commerce, LMS, CMS):** 99.9% uptime.

Disaster Recovery (DR) & Backup

- **Active-Passive Redundancy:** Multi-zone edge deployments ensure frontends remain active during regional CDN outages.
- **Database Backups:** Core configuration data (PIM, CMS schemas) undergo daily snapshot backups with 30-day retention. Customer records in the CRM follow automated Salesforce backup schedules.

Security, Privacy & PCI Scope

- **Data Residency:** Customer PII is stored in databases residing within UK/EU jurisdictions to comply with GDPR/UK-GDPR data sovereignty requirements.
- **PCI-DSS Compliance Scope:** The Frontend Web App and CMS/PIM databases remain entirely out of scope for PCI compliance. Credit card data is collected via hosted elements (iframes/secure scripts) pointing directly to the Payment Gateway.

API Governance

- **Versioning:** All custom service API interfaces must utilize semantic versioning in URI patterns (e.g., `/api/v1/...`).
- **Rate Limiting:** Public API routes are limited to 100 requests per minute per IP address. Downstream systems enforce API token quotas to prevent service exhaustion.